

Sociogenomics

Polygenic scores

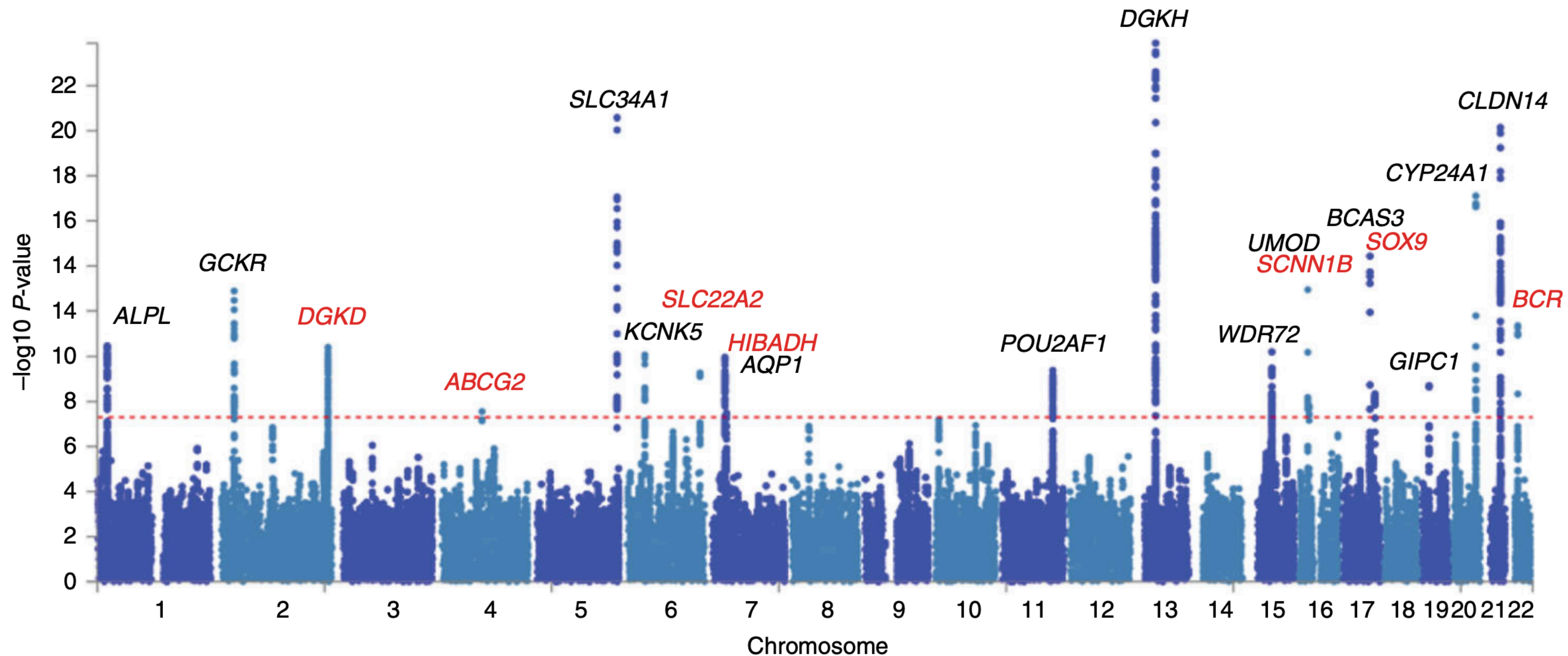
Nicola Barban



Announcements

- This week **Wednesday 17 Aprile. No LAB!**
- **Monday 6 May and Tuesday 7 May NO CLASS!**
- **Guest Lecture by Gaia Ghirardi on Monday 29 April**
- **EXAMS:**
 - **I APPELLO: 20 JUNE 11:00**
 - **II APPELLO: 18 JULY 11:00**
 - **III APPELLO: 17 September 14:00**

gwas results



Individual SNPs are weak predictors

- Weak prediction of individual genetic variants.
- Some exceptions (APOE), FTO
<https://www.snnpedia.com/index.php/Rs1421085>

Abstract

A genome-wide association study (GWAS) of educational attainment was conducted in a discovery sample of 101,069 individuals and a replication sample of 25,490. Three independent single-nucleotide polymorphisms (SNPs) are genome-wide significant (rs9320913, rs11584700, rs4851266), and all three replicate. Estimated effects sizes are small (coefficient of determination $R^2 \approx 0.02\%$), approximately 1 month of schooling per allele. A linear polygenic score from all measured SNPs accounts for $\approx 2\%$ of the variance in both educational attainment and cognitive function. Genes in the region of the loci have previously been associated with health, cognitive, and central nervous system phenotypes, and bioinformatics analyses suggest the involvement of the anterior caudate nucleus. These findings provide promising candidate SNPs for follow-up work, and our effect size estimates can anchor power analyses in social-science genetics.

GWAS of 126,559 Individuals Identifies
Genetic Variants Associated with
Educational Attainment

The 4th law of behavioural genetics

The Fourth Law of Behavior Genetics

[Christopher F. Chabris](#), [James J. Lee](#), [David Cesarini](#), more...

First Published August 12, 2015 | Research Article | [Find in PubMed](#) |

<https://doi.org/10.1177/0963721415580430>

[Article information](#) ▼

1. All human behavioral traits are heritable. [That is, they are affected to some degree by genetic variation.]
2. The effect of being raised in the same family is smaller than the effect of genes.
3. A substantial portion of the variation in complex human behavioral traits is not accounted for by the effects of genes or families.
4. A typical human behavioral trait is associated with very many genetic variants, each of which accounts for a very small percentage of the behavioral variability.

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4. **A typical human behavioral trait is associated with very many genetic variants, each of which accounts for a very small percentage of the behavioral variability.**

Number of Schizophrenia associated SNPs

reason to expect the trend depicted in [Figure 1](#) to continue.

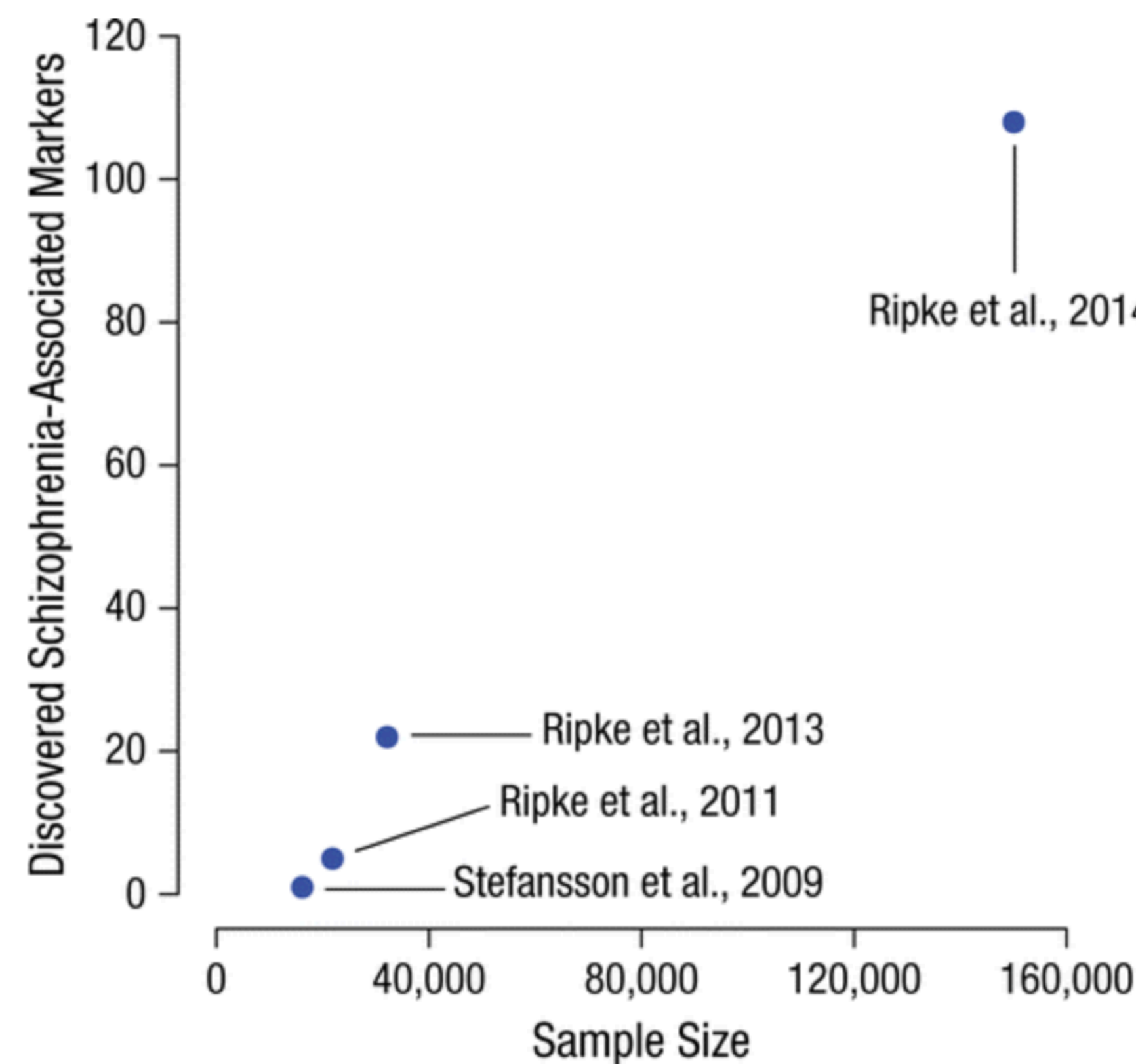
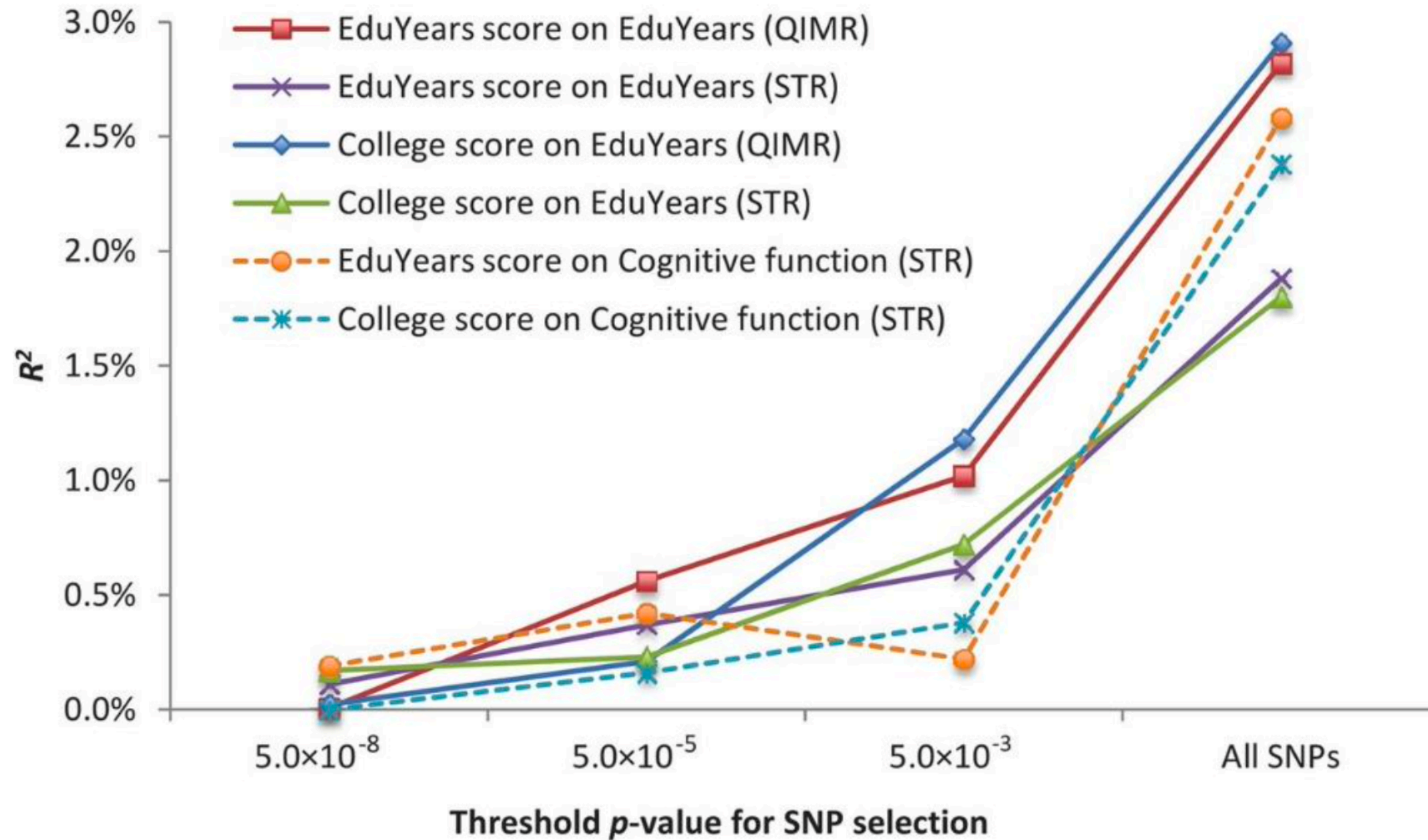


Fig. 1. Number of schizophrenia-associated single-nucleotide polymorphisms clearing the strict genome-wide association study significance threshold ($p < 5 \times 10^{-8}$) as a function of discovery-stage sample size, which has increased over time. Although the four studies presented are not methodologically identical (because of different ratios of cases to controls), the increasing number of “hits” with sample size is nevertheless informative.

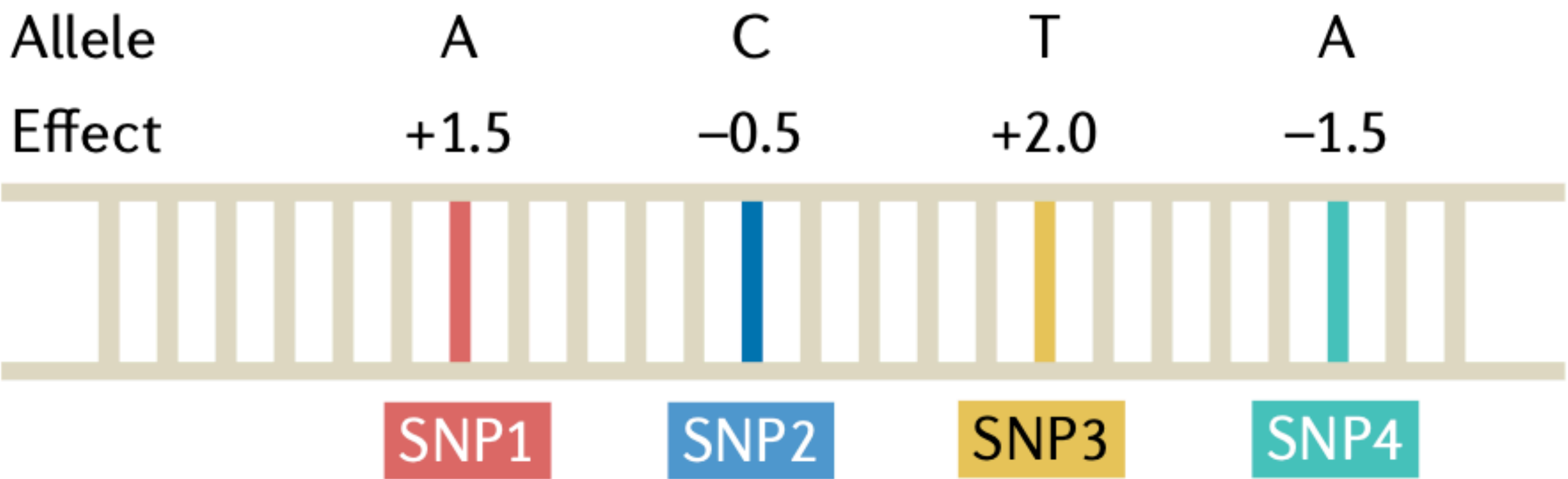
Aggregate R2 from multiple SNPs



Polygenic scores (PGS)

- Also called: Polygenic Risk Scores (PRS) or polygenic Index (PGI)
- Aggregate measures of individual genetic predisposition
- Each individual has a “score” based on her genotype

① GWAS summary statistics



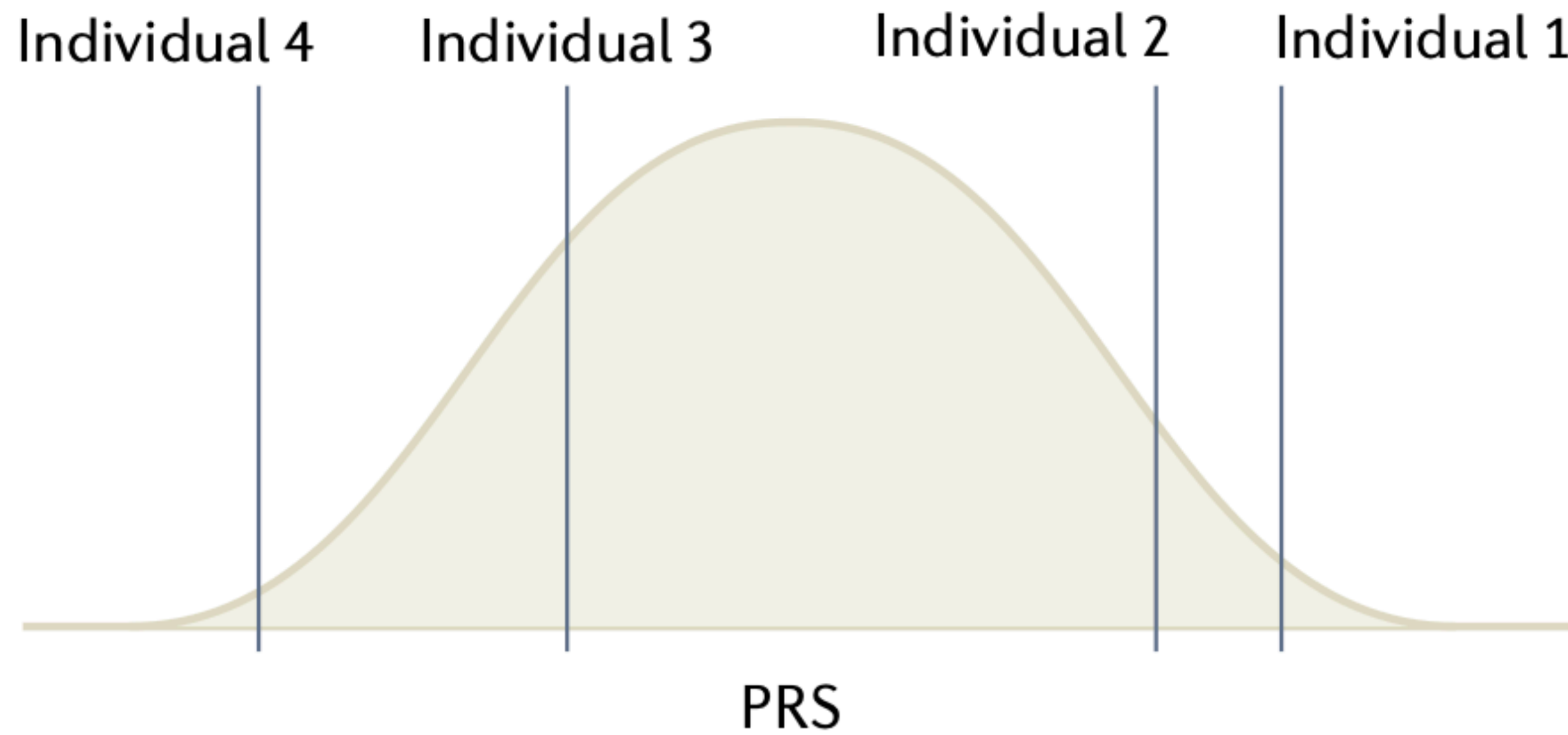
② Genotype data

	SNP1	SNP2	SNP3	SNP4
Individual 1	AT	CG	TT	CC
Individual 2	TA	GG	GT	CA
Individual 3	TT	CC	GT	CA
Individual 4	TT	CC	GG	AA

③ Polygenic risk score

Individual 1	1.5	−	0.5	+	4.0	−	0.0	=	5.0
Individual 2	1.5	−	0.0	+	2.0	−	1.5	=	2.0
Individual 3	0.0	−	1.0	+	2.0	−	1.5	=	−0.5
Individual 4	0.0	−	1.0	+	0.0	−	3.0	=	−4.0

④ PRS distribution



Pros of using polygenic scores

- Identify risk categories across individuals (useful for screening)
- Use as a measure of genetic predisposition in addition to other risk factors
- Use to study Gene-Environment (GxE) interactions



Possible interactions of E with multiple SNPs are too high and need big sample size

Cons of using PGS

- No information on underlying biology
- Dependent on GWAS summary statistics
- Population-specific
- Need to make choices on How many SNPs
- Based on common variants

How to create a polygenic scores

we define a polygenic score for an individual as the **weighted sum** of a person's genotypes at M loci. A PGS for individual i can be calculated as the sum of the allele counts a_{ij} (0, 1, or 2) for each SNP $j = 1, \dots, M$, multiplied by a weight w_j :

$$PGS_i = \sum_{j=1}^J a_{ij} w_j$$

A polygenic score is therefore a **linear combination** of the effects of multiple SNPs on the trait of interest. The underlying model in a PGS is usually additive since we count the number of “risk alleles” for each SNP included in the score,

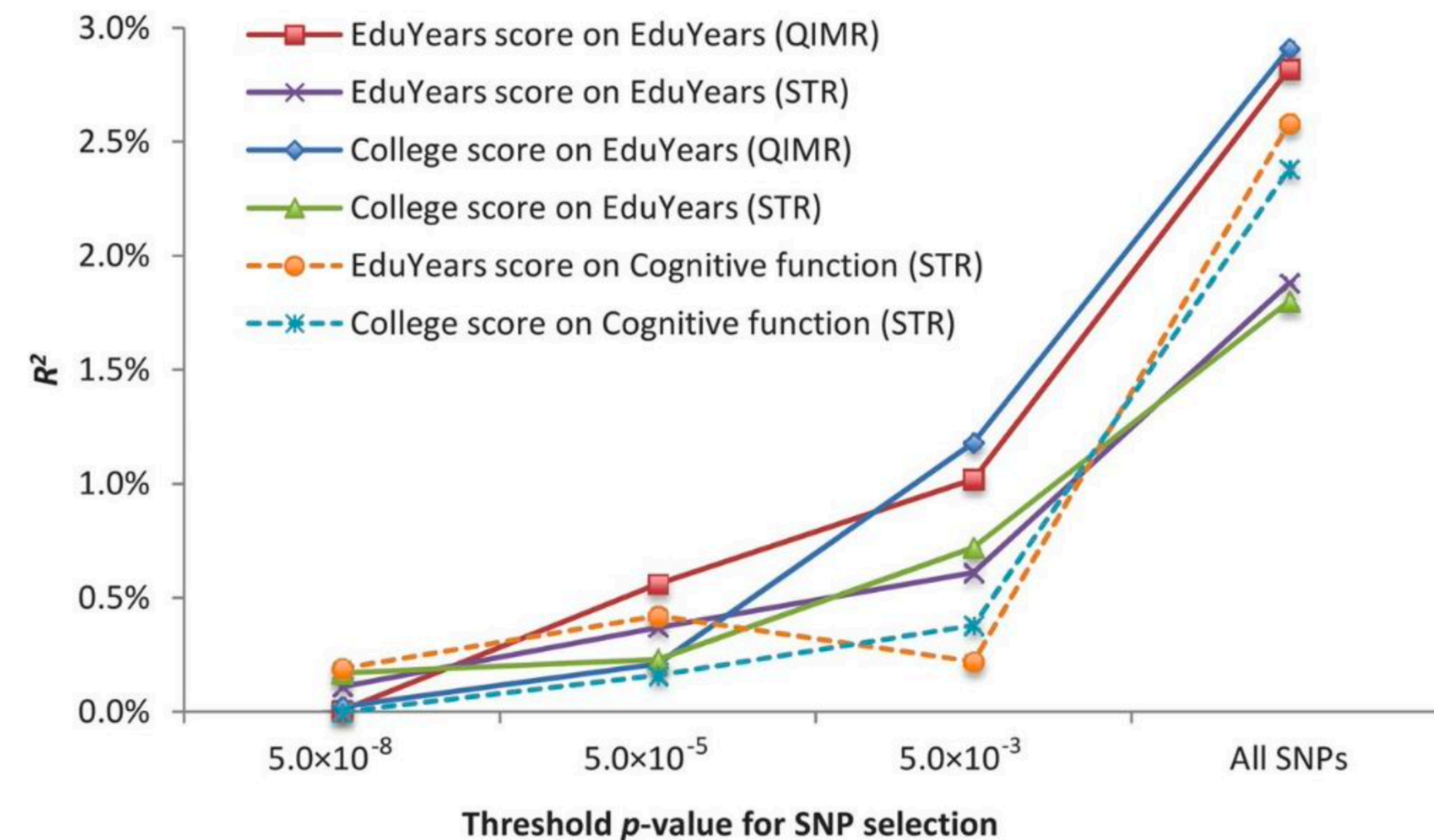
When constructing a PGS, there are two main decisions:

1. **Which (and how many) SNPs** will be used to construct the score?
2. **What weights** will be used?



Pruning & thresholding

SNPs used for the analysis are selected after identifying independent signals (**pruning**). The number of SNPs chosen depends **on p-value thresholds**



Pruning & thresholding

1. Obtain GWAS summary statistics from a large discovery sample.
2. Obtain independent sample(s) that has genome-wide data.
3. Align and select SNPs in common between the two samples.
4. Select independent SNPs by pruning based on linkage disequilibrium.
5. Restrict to SNPs with a p -value that is lower than a certain threshold.
6. Construct PGS by summing up risk alleles weighted by betas from the summary statistics.
7. Evaluate strength of this association by regressing the phenotype on the PGS.

Step 1

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Some example of summary statistics

Table 3 | **Databases of GWAS summary statistics**

Database	Content
GWAS Catalog ¹¹⁰	GWAS summary statistics and GWAS lead SNPs reported in GWAS papers
GeneAtlas ⁸	UK Biobank GWAS summary statistics
Pan UKBB	UK Biobank GWAS summary statistics
GWAS Atlas ²⁷³	Collection of publicly available GWAS summary statistics with follow-up in silico analysis
FinnGen results	GWAS summary statistics released from FinnGen, a project that collected biological samples from many sources in Finland
dbGAP	Public depository of National Institutes of Health-funded genomics data including GWAS summary statistics
OpenGWAS database	GWAS summary data sets
Pheweb.jp	GWAS summary statistics of Biobank Japan and cross-population meta-analyses

For a comprehensive list of genetic data resources, see REF.¹³. GWAS, genome-wide association studies; SNP, single-nucleotide polymorphism.

- **Psychiatric Genomics Consortium** <https://www.med.unc.edu/pgc/download-results/>
- **GIANT Anthropometric** [https://portals.broadinstitute.org/collaboration/giant/index.php/GIANT consortium data files](https://portals.broadinstitute.org/collaboration/giant/index.php/GIANT_consortium_data_files)
- **SSGAC** <http://ssgac.org>

Step 2

1. Obtain GWAS summary statistics from a large discovery sample.
2. **Obtain independent sample(s) that has genome-wide data.**
3. Align and select SNPs in common between the two samples.
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Summary statistics and target sample **MUST** be independent

- If GWAS results are based on the same (or part of) the sample of interest results are likely to be inflated.
- Need to cross validate.
- Sometimes it is a problem with large consortia composed by many cohorts
- Sometimes Large GWAS projects provide ad hoc meta-analysis where specific cohorts are removed from the final results.
- <https://www.thessgac.org/pgi-repository>

Step 3

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SNP alignment

1. This step is to ensure that SNPs are measured consistently in the baseline and target sample.
2. The alleles reported in the summary statistics **(baseline sample)** should thus be the same as in the genotype data.
3. This is not always the case.
 1. The SNPs from the genotype sample **(target sample)** may be different
 2. Different genotyping platforms may report different alleles for the same SNP

SNP alignment

DNA is organized in a double-helix structure composed of four bases: adenine (A), thymine (T), cytosine (C), and guanine (G). Adenine pairs with thymine and cytosine with guanine, forming a sequence organized in two strands such as this:

Chromosome 1

Strand 1: ATCTGG**T**ACTCCAT

Strand 2: TAGACC**A**TGAGGTA

Chromosome 2

Strand 1: ATCTGG**C**ACTCCAT

Strand 2: TAGACC**G**TGAGGTA

GWAS Results

MarkerName	Effect_Allele	Other_Allele	EAF	Beta	SE	Pvalue
rs4075116	T	C	0.78	0.001	0.005	0.8374
rs3934834	T	C	0.13	-0.007	0.005	0.1763
rs3766193	C	G	0.4	-0.006	0.004	0.1371
rs3766192	T	C	0.61	0.007	0.004	0.0978
rs3766191	T	C	0.11	-0.008	0.005	0.133
rs9442371	T	C	0.62	0.006	0.004	0.1567
rs9442372	A	G	0.37	-0.005	0.004	0.2273
rs10907177	A	G	0.87	0.008	0.005	0.1198
rs3737728	A	G	0.26	-0.003	0.004	0.4356

Empirical Data

1 rs3934834 0 1005806 A G

Allele flipping

if a SNP in the base sample is A/C and T/G in the target, we know that they are simply measured on different strands since T is the complement of A and G the complementary base of C. **If this is the case, we need to flip the alleles in the base sample by swapping the A with T or C with G.**

ambiguous SNPS, which are SNPs with the following alleles: A/T and C/G. In this case, it is not possible to say whether the base and target data are referring to the same allele or not

Possible solutions for ambiguous SNPs

- Remove them from the analysis. Close SNPS will capture genetic signals
- Use allele frequencies from a reference panel to flip alleles

Step 4

1. Obtain GWAS summary statistics from a large discovery sample.
2. Obtain independent sample(s) that has genome-wide data.
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4. **Select independent SNPs by pruning based on linkage disequilibrium.**
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7. Evaluate strength of this association by regressing the phenotype on the PGS.

Pruning & Clumping

To avoid the oversampling of a region of the genome that is more densely genotyped, it is recommended to **select only independent SNPs**

This is due to the fact that you avoid basing the PGS on specific regions of the DNA without any substantive reason other than genotype density. In other words, we want to avoid double counting causal variants.

Both **pruning** and **clumping** can be performed.

clumping selects the most statistically significant variant in the locus.

we need to provide some parameters indicating the LD window and the r^2 threshold. The recommended clumping r^2 is between 0.5 and 0.7. An r^2 window that is too small would drop potential causal SNPs, while one that is too large would allow too much “double counting.” The LD window defines the distance after which we suppose that variants are statistically independent. A common choice for the LD window is 1Mb

Step 5

1. Obtain GWAS summary statistics from a large discovery sample.
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5. **Restrict to SNPs with a p-value that is lower than a certain threshold.**
6. Construct PGS by summing up risk alleles weighted by betas from the summary statistics.
7. Evaluate strength of this association by regressing the phenotype on the PGS.

P-value thresholding

how many SNPs do we need to use to construct a PGS? The answer depends on multiple factors. **A common approach is to select SNPs based on the p-value of the association within the summary statistics.** We can decide to select only GWAS significant SNPs (p -value $< 5 \times 10^{-8}$) or at the other extreme, select all SNPs (p -value ≤ 1).

Generally speaking, stricter p -value thresholds are more suitable for traits that are not polygenic while more lenient thresholds perform the best for polygenic traits.

However, the more variants that are included in the calculation, the greater the risk that you include unnecessary “noise” in the PGS. In other words, you are likely to include many variants that are not causal.

Step 6

1. Obtain GWAS summary statistics from a large discovery sample.
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③ Polygenic risk score

Individual 1	1.5	−	0.5	+	4.0	−	0.0	=	5.0
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Step 7

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6. Construct PGS by summing up risk alleles weighted by betas from the summary statistics.
7. **Evaluate strength of this association by regressing the phenotype on the PGS.**

Evaluate predictive power PGS

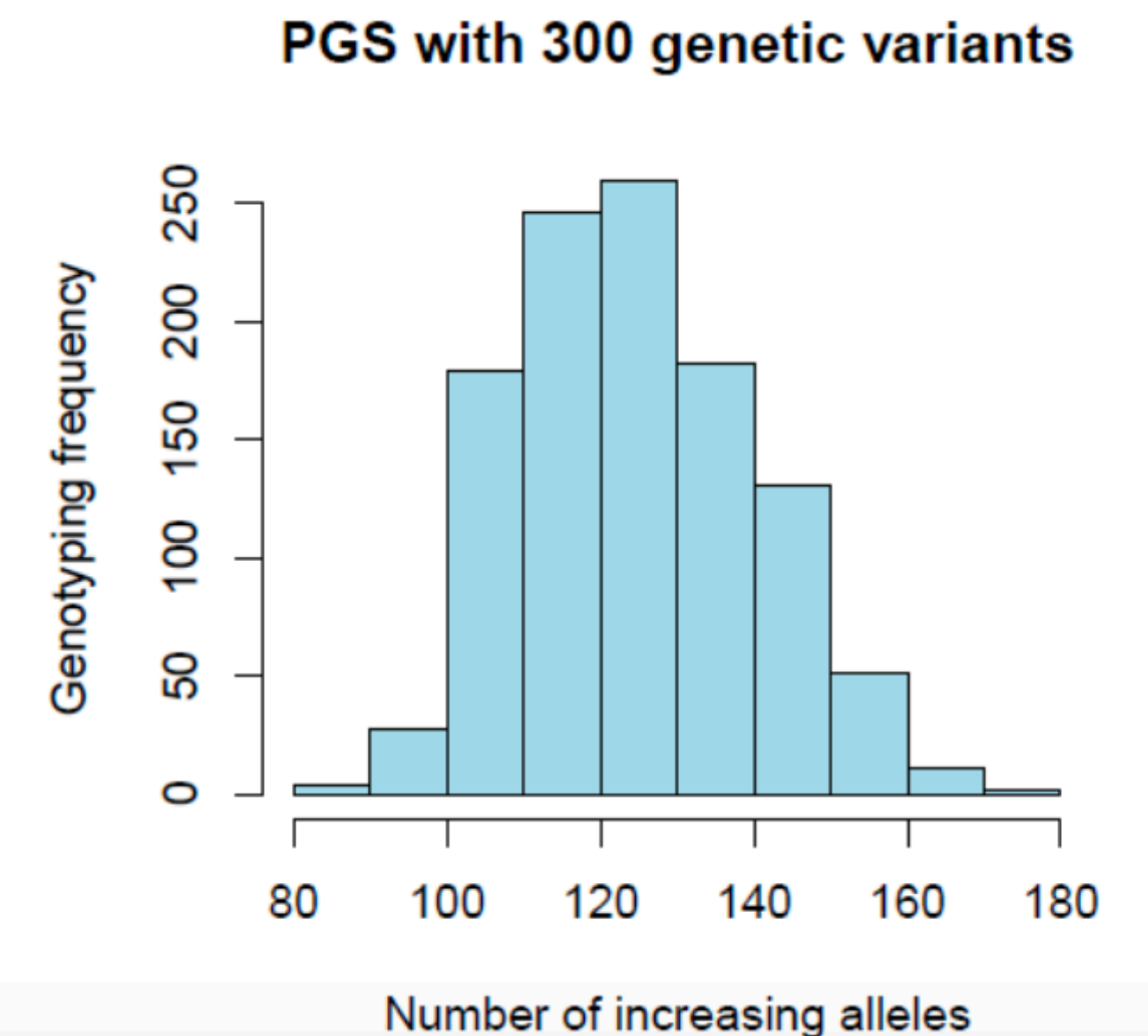
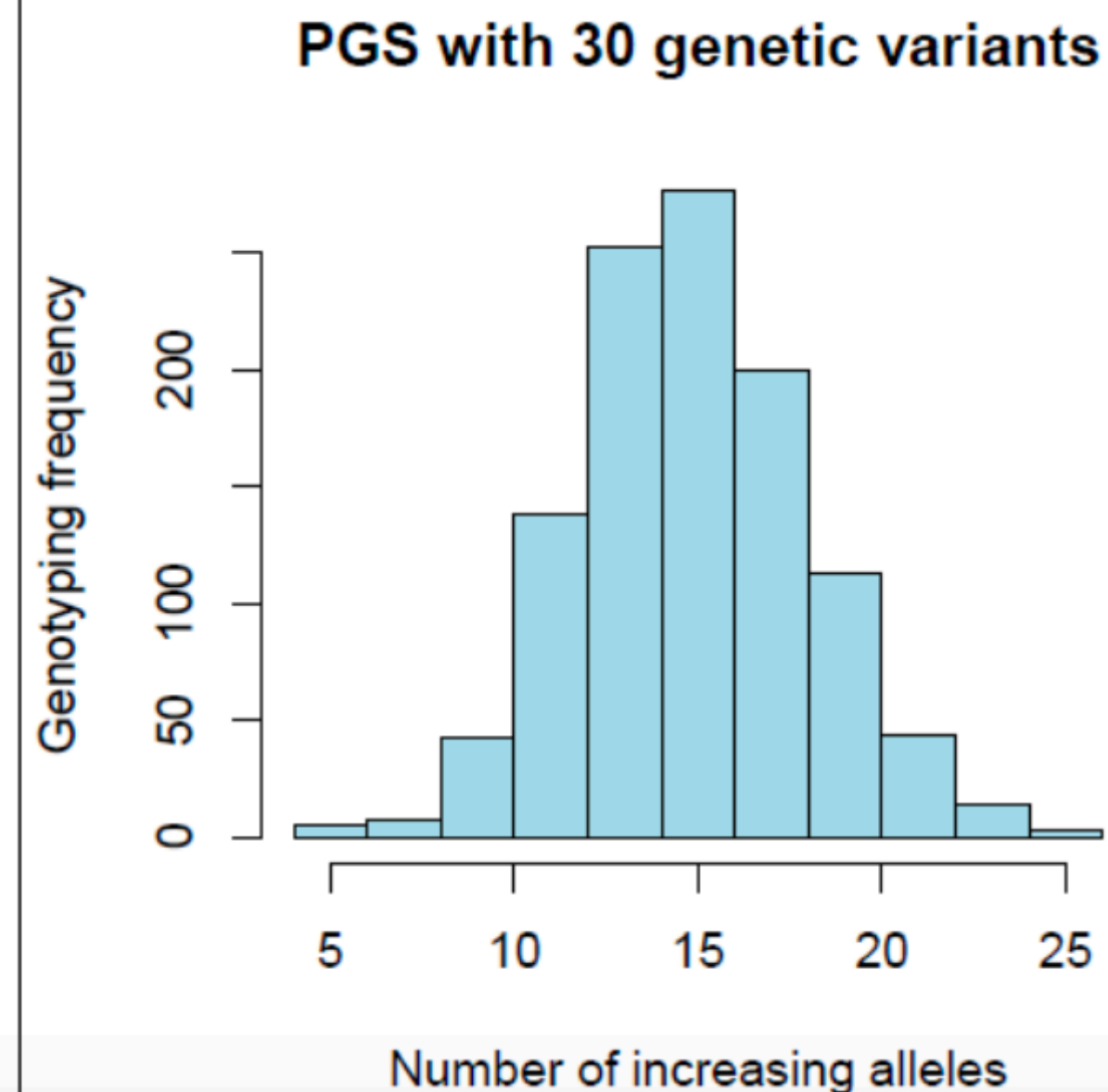
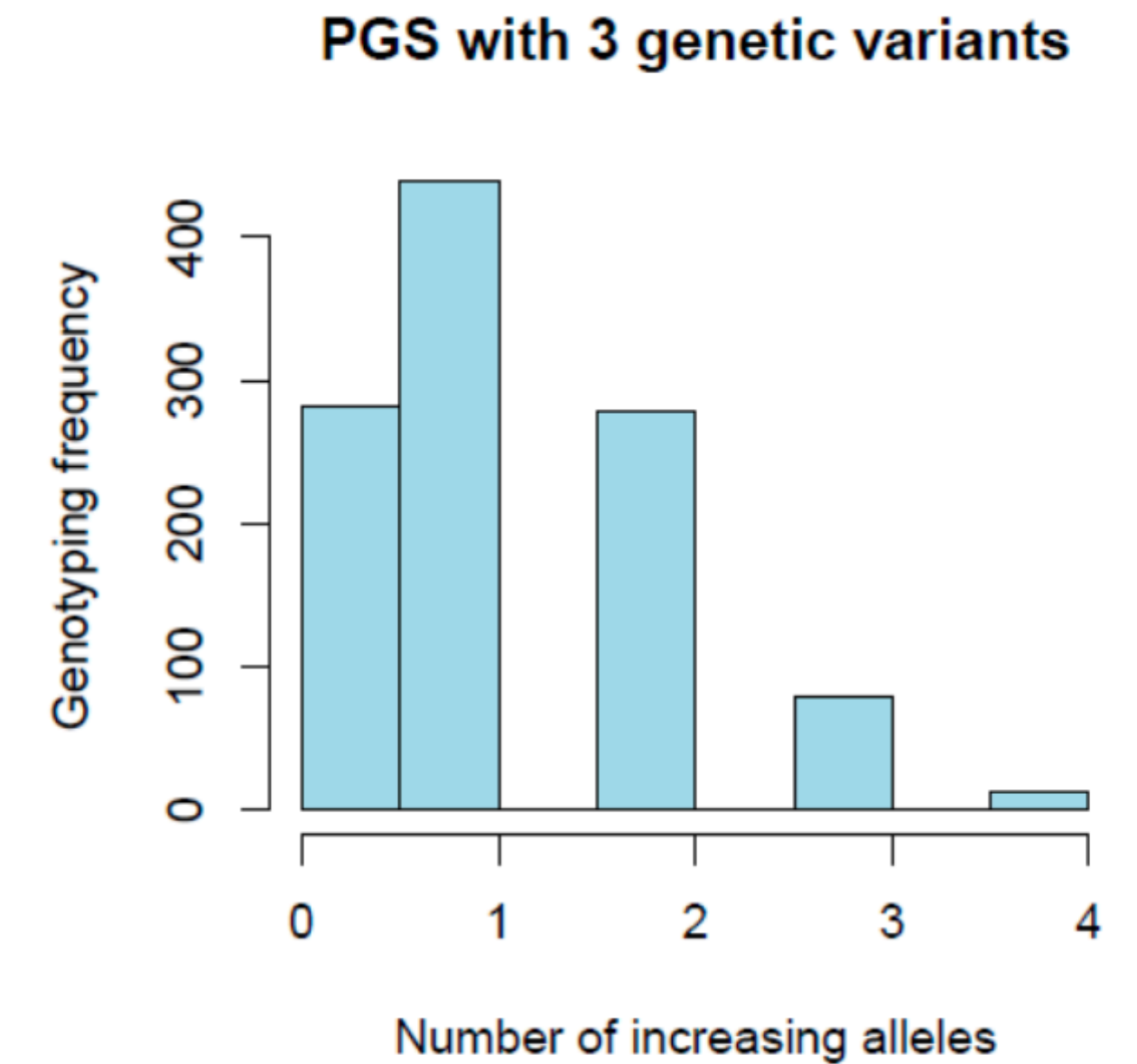
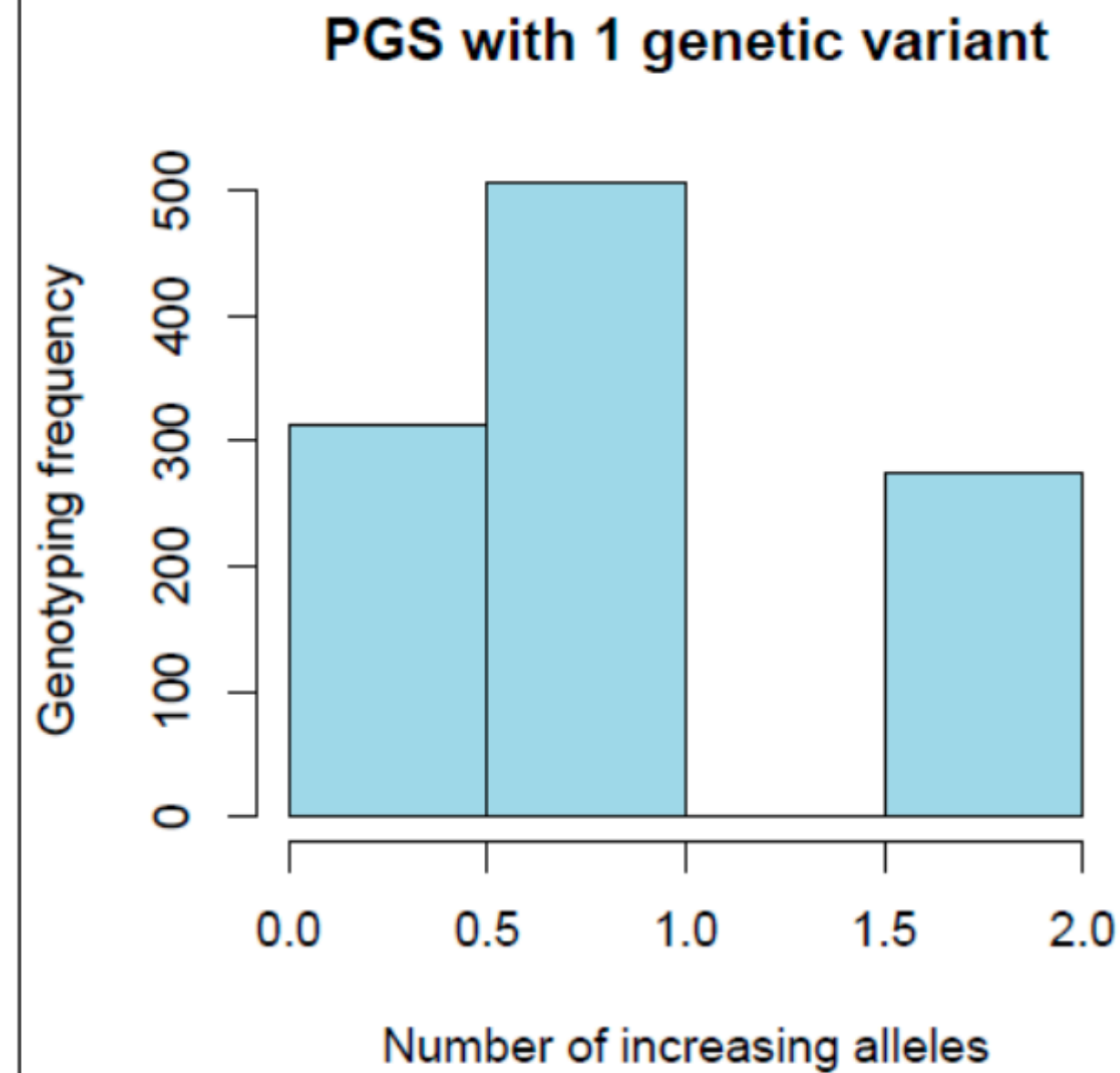
A final and important step is to evaluate the **explanatory or “predictive” power** of a PGS. We note that prediction is a misleading term here since we are actually usually interested in understanding how much variability can be explained by including a PGS in a model

Calculate how much of the variance is explained by the PGS, net of principal components. To do this, we can calculate the **differential in the R²**

It is important, however, to also produce **confidence intervals** on this differential R₂. Because the statistical distribution is not known, it is necessary to nonparametric statistical techniques such as bootstrap. Below we build a small function in R that calculates confidence intervals for the differential R₂.

PGS have normal distribution

Polygenic scores can be thought of as a sum of many independent genetic signals. A central premise of probability theory in statistics, known as the **central limit theorem**, establishes that when many independent random variables are added, their sum tends toward a normal distribution



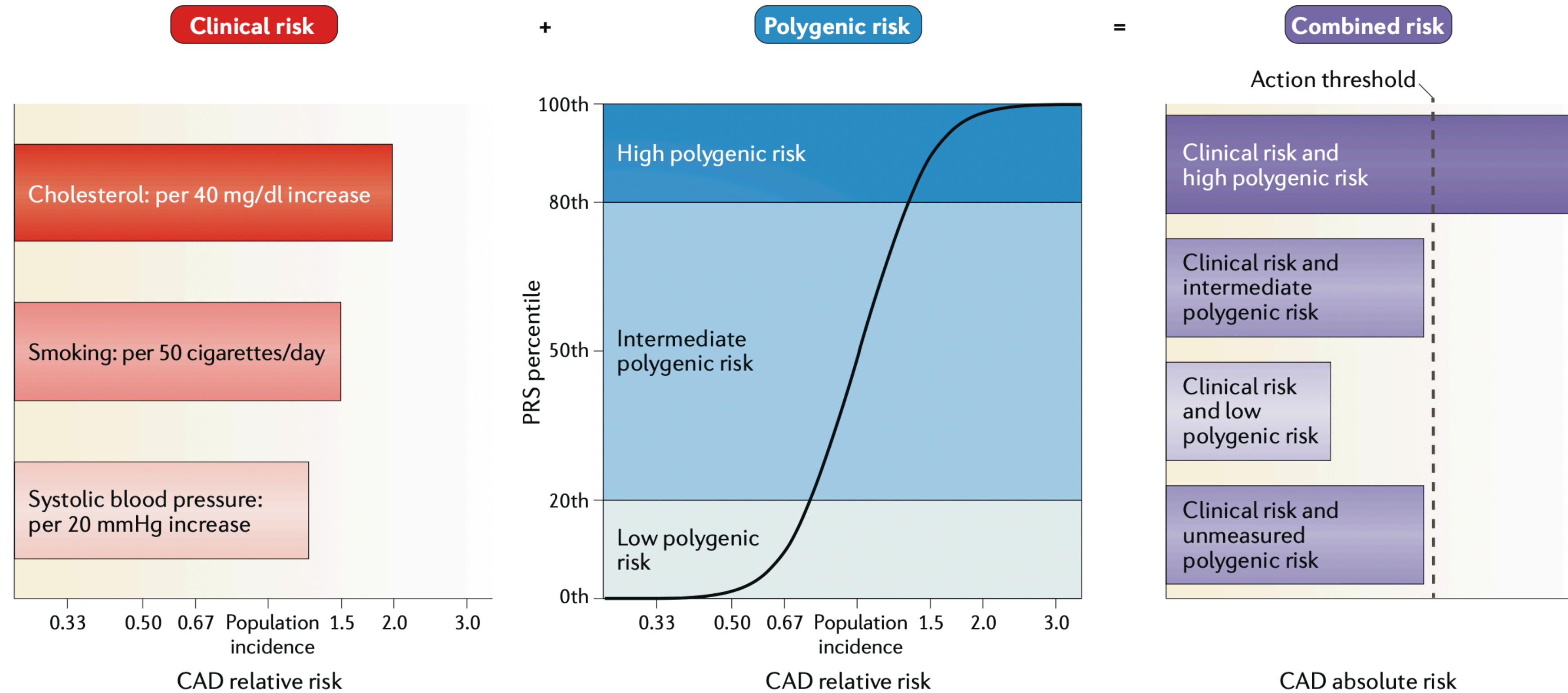


Fig. 1 | **Contrasting and combining clinical risk factors and polygenic risk.** The relative risk conveyed to individuals via commonly measured clinical risk factors (left panel) and polygenic risk estimation (middle panel) for coronary artery disease (CAD) is comparable and, when combined, can lead to different action recommendations (right panel). Relative risks for commonly measured clinical risk factors (left panel) can vary across populations and are approximated here. For polygenic risk (middle panel), the black sigmoidal curve represents the estimated CAD risk relative to average polygenic risk (at population incidence) based on 74 genome-wide association study (GWAS)-significant single nucleotide polymorphisms (SNPs)^{38,39}. The bars represent the percentile thresholds typically used to define low (<20th percentile), medium (20th–80th percentile) and high (>80th percentile) polygenic risk. The combination of clinical and polygenic risk estimates (right panel) can lead to combined risk estimates that exceed the appropriate thresholds of risk versus benefit that justify certain medical interventions (action threshold). In this example, an individual with estimated clinical risk near the action threshold in the absence of polygenic risk information (bottom bar) could clarify their total risk with the addition of a polygenic risk estimate to decide against (low polygenic risk) or for (high polygenic risk) taking clinical action. PRS, polygenic risk score.